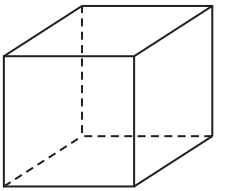
14 The diagram shows a cube on horizontal ground.



$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The cube has sides of length 28.4 cm correct to the nearest mm.

The cube exerts a force of 135 newtons, correct to the nearest 5 newtons, on the ground.

Work out the upper bound for the pressure on the ground due to the cube.
Give your answer in newtons/cm^2
You must show all your working.

Length: $28.35 \leq l < 28.45$ cm

1 mark
1 mm = 0.1 cm;
use half the rounding unit.

Force: $132.5 \leq F < 137.5$ N

Upper pressure = largest force ÷ smallest area

1 mark
Area on the ground is l^2.

1 mark
Use force upper bound
and length lower bound.

$$\frac{137.5}{28.35^2} = 0.1710789483\ldots \text{ N/cm}^2$$

Upper bound ≈ 0.1711 N/cm^2

1 mark
Retain enough figures for the bound.

..... newtons/cm^2

(Total for Question 14 is 4 marks)

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